Notebook III Gate 3B - Corrected Direct Tangent OPE Closure

Carrollian Screening Programme - Notebook III

Gate 3B: Corrected Averaged-Ordered Tangent OPE and Nonlinear Connection Matching

A finite-parent derivation of the A_2/W_3 tangent OPE in which the two ordering orientations, the affine logarithm, the based residue, the barred BCH scalar, and the canonical target frame are kept distinct.

GATE 3B PASS - CORRECTED | GATE 3 CLOSES WITH OUTCOME B | DERIVED ORDERING CONTACT | NO ORDINARY SECOND SCREENING

Corrected final result

For $i = 1, 2$ and $\sigma = \pm$, the coefficient extracted from the two exact finite-parent orientations satisfies

$$\boxed{[Q_i^{[1]}, \mathcal{U}_{3,n}^{\sigma,[0]}]_{\mathrm{PBW}} + \frac{1}{2}\Delta\mathcal{U}_{\mathrm{T},3,n}^{\sigma,[1]} Q_i^{[0]} = 0.}$$

The ordinary commutative Wick diagnostic misses a fixed averaged-order contact. Restoring that derived contact accounts for the previously omitted w^{-2} terms and makes both strict odd OPEs exact total derivatives. There is no ad hoc contact counterterm. $Q_i^{[1]}$ remains a deformation/connection intertwiner, not an ordinary second screening.

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1. Frozen conventions and the corrected total derivative

Write $P_a = J_a^+$, $N_a = J_a^-$, $P_3 = -P_1 - P_2$, $N_3 = -N_1 - N_2$, with

$$N_a(z)P_b(w) \sim \frac{\kappa_{ab}/2}{(z-w)^2}, \qquad [p_a^-, q_b^+] = [p_a^+, q_b^-] = \frac{1}{2}\kappa_{ab}.$$

For $\alpha_1 = (1, -1, 0)$, $\alpha_2 = (0, 1, -1)$, set

$$\begin{aligned} E_i(w) &=: e^{\mu\alpha_i\cdot\varphi^+(w)} :, \\ \alpha^+\mu &= 1, \qquad \nu := \mu_{\mathrm{diff}} = -\alpha^-\mu^2, \\ \mathcal{S}_{i,\mathrm{odd}}^{[1]} &=: (\mu\alpha_i\cdot\varphi^- + \nu\alpha_i\cdot\varphi^+)E_i : . \end{aligned}$$

On the coefficient algebra the ordinary derivative is fixed by

$$\boxed{\partial_w \log(w/w_0) = w^{-1}, \qquad \partial_w(w^{-1}) = -w^{-2}.}$$

The second identity is load-bearing. Omitting it produces the false residual cancellation in the superseded archive.

Cell 1 - corrected inverse-coordinate derivative and residuals

```
import sympy as s
mu,w,P1,P2,P1p,P2p=s.symbols('mu w P1 P2 P1p P2p', nonzero=True)
R1=-mu**2*(mu*P1**2+2*mu*P1*P2-P1p-3*mu*P2**2+P2p)/4
R2= mu**2*(mu*P1**2-P1p-4*mu*P2**2-2*P2p)/4
Re1=-R1-mu**2*(P1+3*P2)/(4*w)-mu/(4*w**2)
Re2=-R2+mu**2*(P1-2*P2)/(4*w)+mu/(4*w**2)
assert s.diff(1/w,w)==-1/w**2
assert s.simplify(R1+Re1+mu**2*(P1+3*P2)/(4*w)+mu/(4*w**2))==0
assert s.simplify(R2+Re2-mu**2*(P1-2*P2)/(4*w)-mu/(4*w**2))==0
print('D(w^-1) and both corrected residuals pass')
```

2. Finite-parent ordering lemma

Let $w(X,a)$ be the parent primary obtained from the differential Miura operator,

$$(a\partial - X_1)(a\partial - X_2)(a\partial - X_3), \qquad w = u_3 - \frac{a}{2}\partial u_2.$$

For $V_\lambda =: e^{\lambda\alpha_i\cdot\phi} :$, a direct parent Wick expansion gives

$$\mathrm{Sing}[w(z)V_\lambda(w)] = \partial_w\left[\left(\frac{B_{i,1}}{z-w} + \frac{B_{i,2}}{(z-w)^2}\right)V_\lambda(w)\right]$$

precisely on the parent screening locus

$$\boxed{a\lambda + \lambda^2 = 1.}$$

The unreduced residual factorizes by $a\lambda + \lambda^2 - 1$ for both roots. This parent identity is reconstructed independently in `independent_check_III_gate3b.py`.

Now substitute the two orientations before taking any OPE coefficient:

$$\begin{array}{ll} \mathrm{u}: & X = P/\varepsilon + \varepsilon N, \quad a = \alpha^+/\varepsilon + \varepsilon\alpha^-, \quad \lambda = \varepsilon\mu_{\mathrm{u}}(t), \\ \mathrm{b}: & X = P/\varepsilon - \varepsilon N, \quad a = \alpha^+/\varepsilon - \varepsilon\alpha^-, \quad \lambda = \varepsilon\mu_{\mathrm{b}}(t), \end{array} \qquad t = \varepsilon^2.$$

The transported barred contour reverses orientation. Consequently it contributes a minus sign to the + contraction and cancels the explicit minus sign in the − contraction. Coefficient extraction from these two parent normal orders is exactly the Fredenhagen-Hoerl averaged order of their equations (4.29)-(4.31). Because the parent OPE has finitely many contractions and poles, coefficient extraction commutes with the singular-part map. This is the required reduction lemma: no polynomial factor is replaced by a commuting surrogate.

The branch equations are

$$\begin{aligned} t\mu_{\mathrm{u}}^2 + (\alpha^+ + t\alpha^-)\mu_{\mathrm{u}} = 1, \qquad & t\mu_{\mathrm{b}}^2 + (\alpha^+ - t\alpha^-)\mu_{\mathrm{b}} = 1, \\ \mu_3 = -\mu^3 - \alpha^-\mu^2, \qquad & \bar\mu_3 = -\mu^3 + \alpha^-\mu^2. \end{aligned}$$

Cell 2 - two finite-parent orientations

```
import sympy as s
e,mu,am=s.symbols('e mu am', nonzero=True)
a,l,X1,X2=s.symbols('a l X1 X2')
P1,P2,N1,N2,Hm,Hp,L=s.symbols('P1 P2 N1 N2 Hm Hp L')
B=[{2:a*l*(a+l)/2,1:-l*(X1*a*a*l+X1*a*l*l-3*X1*a-2*X1*l-X2*a*a*l-X2*a*l*l-X2*a-2*X2*l)/2},
   {2:-a*l*(a+l)/2,1:l*(X1*a*a*l+X1*a*l*l-3*X1*a-2*X1*l+2*X2*a*a*l+2*X2*a*l*l-2*X2*a)/2}]
for i in range(2):
    branches=[]
    for sign in (1,-1):
        m3=-mu**3-sign*am*mu**2
        sub={a:1/(mu*e)+sign*e*am,l:e*(mu+e**2*m3),X1:P1/e+sign*e*N1,X2:P2/e+sign*e*N2}
        F=sign*mu*Hm+m3*Hp+mu**2*L
        branch={}
        for q in (1,2):
            parent=s.series(B[i][q].subs(sub),e,0,4).removeO()
```

```
            coefficient=s.expand(e*parent*(1+e**2*F)/2).coeff(e,2)
            branch[q]=s.factor(coefficient)
        branches.append(branch)
    odd={q:s.factor((branches[0][q]-branches[1][q])/2) for q in (1,2)}
    H=Hm-am*mu*Hp
    target=({2:(H+am)/4,1:(N1+N2+mu*H*(P1+P2))/2},
            {2:-(H+am)/4,1:-(N1+mu*H*P1)/2})[i]
    assert all(s.simplify(odd[q]-target[q])==0 for q in (1,2))
print('both parent orientations reproduce both averaged odd witnesses')
```

3. The (+)-primary OPEs

Theorem 3.1 - unchanged complete singular parts

$$\mathrm{Sing}[\mathcal{U}_3^+(z)\mathcal{S}^{[1]}_{1,\mathrm{odd}}(w)] = \partial_w\left[\left(\frac{P_1+P_2}{2(z-w)} + \frac{1}{4\mu(z-w)^2}\right)E_1\right],$$

$$\mathrm{Sing}[\mathcal{U}_3^+(z)\mathcal{S}^{[1]}_{2,\mathrm{odd}}(w)] = \partial_w\left[\left(-\frac{P_1}{2(z-w)} - \frac{1}{4\mu(z-w)^2}\right)E_2\right].$$

Since $\mathcal{U}_3^+$ contains only commuting P-fields, these two formulas are unchanged by the ordering repair.

4. The audited commutative diagnostic

If the N-containing averaged products in $\mathcal{U}_3^-$ are prematurely replaced by commutative monomials, the strict odd density gives the algebraic residuals

$$R_1 = -\frac{\mu^2}{4}\left[\mu P_1^2 + 2\mu P_1P_2 - P_1' - 3\mu P_2^2 + P_2'\right],$$

$$R_2 = \frac{\mu^2}{4}\left[\mu P_1^2 - P_1' - 4\mu P_2^2 - 2P_2'\right].$$

The common even/affine coefficient

$$\mathcal{S}^{[1]}_{i,\mathrm{even}} =: (-\mu^3\alpha_i\cdot\varphi^+ + \mu^2\ell)\,E_i :, \qquad \ell = \log(w/w_0),$$

has the corrected ordinary residuals

$$\boxed{R^{\mathrm{comm}}_{1,\mathrm{even}} = -R_1 - \frac{\mu^2}{4w}(P_1+3P_2) - \frac{\mu}{4w^2},}$$

$$\boxed{R^{\mathrm{comm}}_{2,\mathrm{even}} = -R_2 + \frac{\mu^2}{4w}(P_1-2P_2) + \frac{\mu}{4w^2}.}$$

Thus the commutative unbarred diagnostic retains

$$\mathcal{R}_1^{\mathrm{comm}} = -\frac{\mu^2}{4w}(P_1+3P_2) - \frac{\mu}{4w^2}, \qquad \mathcal{R}_2^{\mathrm{comm}} = \frac{\mu^2}{4w}(P_1-2P_2) + \frac{\mu}{4w^2}.$$

Correct interpretation

These expressions falsify the superseded cancellation but are not the averaged-order OPE. They diagnose the contact omitted when the two parent orientations are collapsed to commutative multiplication.

5. Derived averaged-order contact and exact strict odd OPE

The two-orientation parent calculation supplies the following correction to the combined unbarred odd-plus-even singular coefficients:

$$\boxed{\alpha_1: \quad \delta\mathcal{O}_{1,3} = 0, \quad \delta\mathcal{O}_{1,2} = \frac{\mu}{4w}, \quad \delta\mathcal{O}_{1,1} = \frac{\mu^2}{2w}(P_1+P_2),}$$

$$\boxed{\alpha_2 : \quad \delta\mathcal{O}_{2,3} = 0, \quad \delta\mathcal{O}_{2,2} = -\frac{\mu}{4w}, \quad \delta\mathcal{O}_{2,1} = -\frac{\mu^2}{2w}P_1.}$$

For example, the root-1 witness shift is $\delta\Omega_{1,1} = \mu/(4w)$. Its derivative contains $-\mu/(4w^2)$; together with the field part of the derived contact it cancels the full $\mathcal{R}_1^{\mathrm{comm}}$. Root 2 works with the opposite sign. This tracks, rather than discards, the two terms identified by the audit.

Define

$$H_i := \alpha_i \cdot \varphi^- - \alpha^- \mu\, \alpha_i \cdot \varphi^+.$$

Theorem 5.1 - complete averaged-order strict odd OPEs

$$\mathrm{Sing}[\mathcal{U}_3^-(z)\mathcal{S}^{[1]}_{i,\mathrm{odd}}(w)]_{\mathrm{Av}} = \partial_w\left[\left(\frac{\Xi_{i,1}}{z-w} + \frac{\Xi_{i,2}}{(z-w)^2}\right)E_i(w)\right],$$

where

$$\boxed{\Xi_{1,2} = \frac{H_1 + \alpha^-}{4}, \qquad \Xi_{1,1} = \frac{N_1 + N_2 + \mu H_1(P_1+P_2)}{2},}$$

$$\boxed{\Xi_{2,2} = -\frac{H_2 + \alpha^-}{4}, \qquad \Xi_{2,1} = -\frac{N_1 + \mu H_2 P_1}{2}.}$$

Equivalently, the complete poles are

$$\begin{aligned}
\mathcal{O}^{\mathrm{Av}}_{1,3} &= \frac{H_1 + \alpha^-}{2},\\
\mathcal{O}^{\mathrm{Av}}_{1,2} &= \frac{\mu H_1(3P_1 + P_2) + 3N_1 + N_2}{4},\\
\mathcal{O}^{\mathrm{Av}}_{1,1} &= \frac{1}{2}[\mu^2 H_1(P_1^2 - P_2^2) + \mu H_1(P_1' + P_2') + 2\mu\,\mathrm{Av}(N_1P_1 - N_2P_2) + N_1' + N_2' - \alpha^-\mu^2(P_1^2 - P_2^2)],\\
\mathcal{O}^{\mathrm{Av}}_{2,3} &= -\frac{H_2 + \alpha^-}{2},\\
\mathcal{O}^{\mathrm{Av}}_{2,2} &= -\frac{\mu H_2(3P_1 + 2P_2) + 3N_1 + 2N_2}{4},\\
\mathcal{O}^{\mathrm{Av}}_{2,1} &= -\frac{1}{2}[\mu^2 H_2(P_1^2 + 2P_1P_2) + \mu H_2 P_1' + 2\mu\,\mathrm{Av}(N_1P_1 + N_1P_2 + N_2P_1) + N_1' - \alpha^-\mu^2(P_1^2 + 2P_1P_2)].
\end{aligned}$$

No remainder survives. The even coefficient is independently exact. With $K_i = \ell - \mu\alpha_i \cdot \varphi^+$, its witnesses are

$$\begin{aligned}
(\Xi^{\mathrm{even}}_{1,1}, \Xi^{\mathrm{even}}_{1,2}) &= \left(\frac{\mu^2 K_1(P_1+P_2)}{2}, \frac{\mu K_1}{4}\right),\\
(\Xi^{\mathrm{even}}_{2,1}, \Xi^{\mathrm{even}}_{2,2}) &= \left(-\frac{\mu^2 K_2 P_1}{2}, -\frac{\mu K_2}{4}\right).
\end{aligned}$$

The old statement that the even channel cancels R_i is therefore replaced: the finite-parent average first restores the ordering contact, after which both odd and even coefficients are separately total derivatives.

6. Exact horizontal boundary

Let $\lambda_i^{\pm}(t) = \alpha_i \cdot \lambda^{\pm}(t)$ denote the root component of the source momentum. The exact horizontal section is

$$\lambda_i^+ = \frac{1}{2}\left(\frac{m - t\mu_{\mathrm{u}}^2}{\mu_{\mathrm{u}}} + \frac{m - t\mu_{\mathrm{b}}^2}{\mu_{\mathrm{b}}}\right), \qquad t\lambda_i^- = \frac{1}{2}\left(\frac{m - t\mu_{\mathrm{u}}^2}{\mu_{\mathrm{u}}} - \frac{m - t\mu_{\mathrm{b}}^2}{\mu_{\mathrm{b}}}\right).$$

It solves separately

$$\begin{aligned}
a_{\mathrm{u}} &= t\mu_{\mathrm{u}}^2 + \mu_{\mathrm{u}}\lambda_i^+ + t\mu_{\mathrm{u}}\lambda_i^- - m = 0,\\
a_{\mathrm{b}} &= t\mu_{\mathrm{b}}^2 + \mu_{\mathrm{b}}\lambda_i^+ - t\mu_{\mathrm{b}}\lambda_i^- - m = 0.
\end{aligned}$$

For a weight-one parent field the based boundary is

$$\mathrm{LRes}_{w_0}\,\partial_w(w^{n+1}V) = \frac{e^{2\pi i a}-1}{2\pi i}\,w_0^{n+1}V(w_0).$$

Since both a_{u} and a_{b} vanish as exact formal series, each parent boundary vanishes before taking its coefficient. The odd half-difference therefore obeys

$$[\widetilde{Q}_i^{[1]}, \mathcal{U}_{3,n}^{\sigma,[0]}]_{\mathrm{moving}} = 0.$$

Cell 3 - exact branch monodromies

```
import sympy as s
t,m,u,b=s.symbols('t m u b', nonzero=True)
du,db=t*u**2,t*b**2
lp=s.Rational(1,2)*((m-du)/u+(m-db)/b)
tlm=s.Rational(1,2)*((m-du)/u-(m-db)/b)
au=s.simplify(du+u*lp+u*tlm-m)
ab=s.simplify(db+b*lp-b*tlm-m)
assert au==0 and ab==0
a=s.symbols('a')
B=(s.exp(2*s.pi*s.I*a)-1)/(2*s.pi*s.I)
assert B.subs(a,au)==0 and B.subs(a,ab)==0
print('both exact parent boundaries vanish before coefficient extraction')
```

7. Barred BCH subtraction

For $(\alpha_i,\alpha_i)=2$, conversion of the transported barred anti-normal vertex to the common normal order gives

$$\bar{Z}_{t,\rho} = (1-\rho)^{-2t\bar{\mu}(t)^2}.$$

This factor is scalar and insertion-coordinate independent. Minimal subtraction removes its divergence. A finite subtraction changes $Q_i^{[1]}$ only by $c_i Q_i^{[0]}$, which does not alter the four equations because the leading charge screens both leading primaries.

8. Canonical PBW target frame

The raw coefficient maps between moving target fibres whose relative tangent is

$$\Delta\mathbf{p}_i^{+} = \mu\alpha_i, \qquad \Delta\mathbf{p}_i^{-} = \nu\alpha_i.$$

With

$$\mathsf{A}_{\mathrm{T},i} = \mu\alpha_i\cdot q^{-} + \nu\alpha_i\cdot q^{+}, \qquad \boxed{Q_i^{[1]} = \widetilde{Q}_i^{[1]} - \mathsf{A}_{\mathrm{T},i}Q_i^{[0]}},$$

the crossed Weyl brackets and the frozen Gate-3A identity $\mathfrak{d}_i^{-}\mathcal{U}_3^{-} = \mathcal{A}_i$ give

$$[\mathcal{U}_3^{+}, \mathsf{A}_{\mathrm{T},i}] = \frac{\mu}{2}\mathcal{A}_i, \qquad [\mathcal{U}_3^{-}, \mathsf{A}_{\mathrm{T},i}] = \frac{1}{2}(\mu\mathcal{B}_i + \nu\mathcal{A}_i).$$

Theorem 8.1 - nonlinear target-frame matching

$$\boxed{[Q_i^{[1]}, \mathcal{U}_{3,n}^{+,[0]}]_{\mathrm{PBW}} = -\frac{\mu}{2}(\mathcal{A}_i)_n Q_i^{[0]},}$$

$$\boxed{[Q_i^{[1]}, \mathcal{U}_{3,n}^{-,[0]}]_{\mathrm{PBW}} = -\frac{1}{2}(\mu\mathcal{B}_i + \nu\mathcal{A}_i)_n Q_i^{[0]}.}$$

These are exactly $-\frac{1}{2}\Delta\mathcal{U}_{\mathrm{T},3}^{\sigma,[1]}Q_i^{[0]}$. The sign comes from the target-frame derivative and the factor $1/2$ from the crossed Weyl bracket.

Cell 4 - PBW sign and one-half factor

```
import sympy as s
U=s.Matrix([[2,0],[0,-1]])
A=s.Matrix([[0,3],[0,0]])
Q0=s.eye(2)
dU=2*(U*A-A*U)
Qpbw=-A*Q0
assert U*Qpbw-Qpbw*U == -dU*Q0/2
print('target-first PBW sign and one-half factor pass')
```

9. Gate disposition

- Correct derivative $D(w^{-1}) = -w^{-2}$ - PASS. Witness: Section 1 and verifier.
- Audited ordinary residuals - PASS. Witness: Section 4.
- Two finite-parent ordering orientations - PASS. Witness: Section 2 and independent reconstruction.
- Derived averaged-order contact - PASS. Witness: Section 5.
- Four complete strict odd OPEs - PASS. Witness: Theorems 3.1 and 5.1.
- Exact horizontal branch boundaries - PASS. Witness: Section 6.
- Barred BCH scalar subtraction - PASS. Witness: Section 7.
- Nonlinear target-frame commutators - PASS. Witness: Theorem 8.1.
- Ad hoc contact counterterm - NONE. Witness: The contact is forced by parent ordering.
- Gate 3B - PASS. Witness: Corrected Outcome B.
- Gate 3 - CLOSED. Witness: Leading theorem plus corrected tangent equation.

10. Claim boundary and references

Authorized conclusion

Gate 3 closes only in the deformation-intertwiner sense. The calculation does not establish an ordinary second screening, a quantum screening doublet, kernel equality, a BRST complex, a screening-screening algebra, or the bulk normalization match.

The next programme step is Notebook IV: semiclassical bulk normalization and the $N = 2$ anchors, after independent freeze review of this corrected Gate-3B archive.

[**FH26**] S. Fredenhagen and L. Hoerl, *Free-field construction of Carrollian W_N-algebras*, JHEP 02 (2026) 218, arXiv:2509.14441v3. Used for the contracted Miura fields and the symmetric averaged-ordering prescription, especially Section 4.4 and equations (4.29)-(4.31). Equations (4.18)-(4.21) are free-field formulas, not a screening construction.

[**FL88**] V. A. Fateev and S. L. Lukyanov, *The models of two-dimensional conformal quantum field theory with Z_n symmetry*, Int. J. Mod. Phys. A 3 (1988) 507.

[**BS93**] P. Bouwknegt and K. Schoutens, *W-symmetry in conformal field theory*, Phys. Rept. 223 (1993) 183-276, arXiv:hep-th/9210010.

[**HLZ11**] Y.-Z. Huang, J. Lepowsky and L. Zhang, *Logarithmic tensor category theory for generalized modules for a conformal vertex algebra, I-II*, arXiv:1012.4193 and arXiv:1110.1929. Background only; the based logarithmic residue is defined in frozen Notebook II.